Binary from the paper we were given:
Extracted from the paper using text extraction from photo on phone
1000110101010010100000100000110000111101000100000011000101100
0110111000010111001101000110111010111001110010111001010111001100110 1
1001001000000101101010000011100001101111111001011101000100000 0
1100100110001010000001011010100000111010111001111001011110010010
0000011010011110011010000011100111101111110110001000000101101010
0000111000011000011110011111001101000001101001111001101000001101
1001101111111001110110001011000101100010110001011001010000001011 0
1010000011101111010001100101111001011001010100000101100010110000
1011000101100001000001101001111001101000001100001010000010101
0011010011101101110010110011001101111110001111010110100000110001111 0
1111110010011001010100000110000111011101100100010000010110010100
0000110100111100110100000111010011010001100101010000011011001100
0001111001111101000100000110001111010001100001111001011000011100
0111110100110010111100100100000110111111001100100000111010011010 0
0110010101000001100110110100111011101100001110110001000000101010
0110100111011011100101100110011011111100011110101101000001101000110
0001111001111010000100000010110101000001010100110100111011011110
0101100110011011111100011110101101000001100101111000011010111110001111
0100001000001101001111001101000000001010010000001000000110
0100110000011001001101100101101011000001100010101101011000000 11
0011000001000000101101010000011100111111001110111011000110100000
0111100111011111110101111001001000001110100110100111011011100101010 0
0001110101111001111010011101110110011101000001110100110010111011001 1
0111011001011110100010000011011111110010010000011011101100101111010
0110001111000011110100010000011011111101110010000011100001101111111
0010111010001000000110101011010001100110110010011000100010100

This version is slightly off, so we used the version from canvas.
Doing the same thing to the document from canvas it became:

```
FTP at 138.47.99.32 - port 21 - user is sol - pass is losXXXXY - where XXXX is a
TimeLock code and Y is the last character of the final TimeLock hash - TimeLock epoch
is
 2026-01-01 00:00:00 - sync your time using telnet or netcat on port 54321
```

We were able to log into the ftp server at 11:47 with password losee10f and retrieve its files.
The file beginning with 7e98cdde that had jupiter on it showed signs of steg encryption.
`
./steg -r -B -o 0 -i 32 -w hidden.bmp > extracted
Found
0100000100100000011000110110100101110000011010000110010101110010001000000111010
0001101000011000010111010000100000011100110111010001100001011100100111010001111
0011001000000111011101101001011101000110100000100000011000010010000001010110000
1011100010000000100000010101010111001101100101001000000110100101110100011100110
0010000001101110011000010110110101100101001000000110000101110011001000000110000
0100100000011010110110010101111001001000000111010001101111001000000110011101110
0101011010010110010001100101001000000111100101101111011101010010000001110100010
1010000111001001101111011101010110011101101000001000000111010001101000011001010
0010000001110011011101000110000101110010011100110010000100001010010000100111010
1100100000011010000110010101110000011011110010000001101011011100110010000000110
1111011100000110101100100000011001100111001001101011011110100111001001101001011
1000101110100011010110010000001100010011010100010000001101011011011000110010000
1100001001000000110100101101100011011100111000001100011011010010110100101101111
1011010110010000001100001011101100111100001111001001000000111100100111010011000
1110110001101110100011001010010000001100011001110100110010101100101011110010110
0011100101110001000000010000001000010011011100110100101100001001000000110100000
0111011001100111011010100110110001101011001000000111100001110101011010010010000
0001101010011110000111011001111010011110010010111000001010
Binary was put into
ASCII Code Converter - Online Hex/Binary to Text Decoder, Translator
to get:

```
A cipher that starts with a V.  Use its name as a key to guide you through the stars!
Bw hepo ks opk frkzriqtk bj klda ilnpciiok avxy y:gcte c:eeyg.  Bnia hvgjlk xui jxvzy.
```

Decrypting the text with the key “Vigenere” gave:

- Go back to the beginning of this challenge with u:luna p:anul. Then decode the stars

We then ftp’d back into 138.47.99.32 with those credentials and retrieved 2 files

Retrieved files:
3d2fad84f4e4a738dff8c58af7bf0742
&
740cfd3199e2a52a2331528145eab143

One file was renamed “key” one was renamed “encrypted” then they were Xor’d

After changing the output file to .bmp, this image was displayed

Go back to the beginning of this challenge with u:nebula p:aluben

After ftping into the server with those credentials, we were able to download the “conclusion” file with this inside:

Name the first human to walk on the Moon. I must know! Please tell me!
Neil Armstrong

Contributions

Saurav Shah:

- Analyzed the 10 bmp images retrieved from the ftp server and identified the file containing the hidden information.
- Extracted the hidden text using byte mode steg and their correct setting
- Used vigenere decryption to recover the next ftp username and password.

Hayden Mersky

- FTP’d into server with u: sol p: los ee10f
- Downloaded all image files from FTP server
- XOR’d files from “luna” login
- Converted to BMP to receive “back to the beginning” message
- Downloaded conclusion file and answered final question

Andy Milletello

- Decoded the initial ascii
- Found the settings needed for the first steg
- Retrieved the ascii in the first steg output
- Found the ascii could be vigenered using vigenere key
- got the 2 files from the second ftp

Amanda Berg

- Helped look for decoding of initial binary text
- Got the password for the ftp server using the timelock program and logged into FTP server
- Downloaded the images from the FTP server

- Helped look for the correct parameters for steg program on the image of jupiter

Ryan Bourge

- Logged into ftp server
- Changed setting on the steg program to help find the correct offset and interval for the given image